Treating diseases

9C Threat from disease · Lesson 4 of 7

Date _____ Class _____

Today I can

- ☐ I can explain which medicines treat which types of disease, and why antibiotics do not work on viruses.
- ☐ I can explain how large amounts of human hormones are made quickly using genetically modified bacteria in a fermenter.

What happens today

1. Do Now: which treatment?
2. Learn: Drugs and medicines · Diseases we cannot cure · Transgenic bacteria · and more
3. Core tasks
4. Challenge, then the exit ticket

Student Book pages 44 to 45 · Workbook pages 32

What I learned today (one or two sentences)

Key Vocabulary

Term	Definition
drug	a substance that affects how the body works
medicine	a drug used to treat or prevent disease
antibiotic	a medicine that kills bacteria
antiviral	a medicine that stops viruses replicating (copying themselves)
genetic disorder	a disease caused by faulty genes that a person is born with
autoimmune disease	a disease in which white blood cells attack the body's own cells
insulin	a hormone made in the pancreas that lowers blood glucose
transgenic	contains DNA from more than one species
fermenter	a large container in which microorganisms are grown
sterilise	kill all the microorganisms in or on something
aseptic	free from unwanted microorganisms

Quick Review

which treatment?

1 An ear infection is caused by bacteria. Would you give an antibiotic or an antiviral?

2 Scurvy is caused by too little vitamin C. How would you treat it?

3 A vaccine prevents a disease. Can it treat someone who already has it?

Core 1 of 2

Everyone does these. Answer on the card or under it.

1 Match each scientific word to its definition. Draw a line.

medicines	drugs that prevent viruses replicating
antibiotics	a hormone that makes cells take up glucose from the blood
antivirals	drugs used to treat or prevent disease
vaccines	substances that make us immune to particular diseases
insulin	drugs that kill bacteria

2 Tick one box in each row to show the best treatment for each disease.

Disease	antibiotic	antiviral	add a nutrient	inject a hormone	vaccine (prevents it)
ear infection caused by bacteria					
flu (caused by a virus)					
scurvy (too little vitamin C)					
type 1 diabetes (no insulin made)					
measles, before a child catches it					

Core 2 of 2

Everyone does these. Answer on the card or under it.

3 Complete the text about fermenters using words from the word bank.

Transgenic bacteria are grown in a large container called a _____. Before use it is _____ with steam so that no other _____ can grow inside it. _____ air is bubbled in so that the bacteria can respire. _____ paddles mix the bacteria, nutrients and oxygen, and a water _____ keeps the temperature steady.

Challenge 1 of 2

When Core is finished. Full sentences.

C1 Transgenic bacteria are grown in fermenters.

a) Give the reason why air is bubbled into the fermenter.

b) Describe two ways in which the fermenter keeps the bacteria growing at a high rate.

Challenge 2 of 2

When Core is finished. Full sentences.

C1 *continued*

c) Describe the aseptic precautions taken when preparing the fermenter, and explain why they are needed.

C2 Scientists can also remove genes from an organism. Explain how this can create a genetically modified organism that is not transgenic.

Exit Ticket

Hand this card in at the end of the lesson.

Name _____

1 What is the difference between an antibiotic and an antiviral?

2 What does 'transgenic' mean?

3 Give one reason why a fermenter is sterilised before use.

Today I can: Goal 1 ☐ Goal 2 ☐